

ARYAN RANE

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Professional Summary

Information Technology student with hands-on experience in Full Stack Development, Cloud Computing, DevOps, Machine Learning, and Data Analytics. Skilled in React.js, Node.js, Python, AWS, Docker, MongoDB, and Power BI, with experience building scalable web applications and cloud-native solutions through academic and personal projects.

Education

Shah & Anchor Kutchhi Engineering College

BTech – Information Technology

GPA: 8.0+ / 10.0

Mumbai, India

Currently in 3rd Year

Technical Skills

Languages	JavaScript, Python, C++, C
Frontend	React.js, Tailwind CSS, Vite, JavaScript
Backend	Node.js, Express.js, MongoDB, MySQL, JWT Auth, REST APIs
Cloud & DevOps	AWS (Lambda, S3, API Gateway), Docker, Jenkins, Prometheus, Grafana, Git, Vercel
ML & NLP	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, TF-IDF, Naive Bayes
Data & Analytics	Power BI (Advanced), Plotly
Security	AES-256 Encryption, Cryptography Fundamentals, Computer Networks

Projects

Legal Rights Awareness Portal *React 18, Node.js, Express.js, MongoDB, Docker, Jenkins, Selenium, Prometheus, Grafana*
— *Live on Vercel*

- Built a full-stack legal education portal featuring 8+ learning modules, interactive quizzes, flashcards, JWT authentication, Dockerized deployment, and CI/CD automation.
- Implemented Docker containerisation, Jenkins CI/CD, Selenium automated testing, and Prometheus + Grafana monitoring; managed sprints via Jira.
- Aligned with SDG 16 (Justice & Strong Institutions) to promote legal awareness among students.

AegisNet AI – Enterprise Network Intrusion Detection Dashboard *React 18, TypeScript, Vite, Tailwind CSS, D3.js, Recharts, Framer Motion, Supabase, Chrome MV3*

- Built a real-time AI-powered intrusion detection dashboard using an Isolation Forest ML model to score every network packet, with XAI forensics for per-packet feature importance.
- Implemented NLP-based threat classification using TF-IDF and Naive Bayes for browser-native attack pattern detection, with no external API calls.
- Includes a Chrome Extension (MV3) for live browser traffic capture, MITRE ATT&CK mapping, CVE references, attack replay for 7 attack types, and one-click firewall mitigation.
- Collaborated with Dhwananya Pachupate and Sakshi Goswami on the project's development.

CloudFrame – Serverless Image Upload & Processing System *AWS Lambda, Amazon S3, API Gateway, Node.js, IAM, HTML5, CSS3, JavaScript*

- Designed and deployed a fully serverless, event-driven image processing pipeline on AWS – uploads via pre-signed S3 PUT URLs trigger Lambda functions that auto-resize images and generate WebP thumbnails.
- Achieved 100% test case pass rate across all 8 scenarios; warm Lambda invocations averaged 2.1s. Enforced least-privilege IAM, 5-minute pre-signed URL TTL, and 7-day S3 lifecycle expiry.

SentiScope – Twitter (X) Sentiment Analysis *Python, Flask, scikit-learn, TF-IDF, Logistic Regression, joblib, Chart.js, Pandas*

- Trained a scikit-learn Logistic Regression + TF-IDF pipeline on 50,000 tweets (Sentiment140) with bigram feature extraction, L2 regularisation, and model serialisation via joblib.
- Deployed as a Flask web app returning real-time sentiment predictions with per-class confidence scores visualised via Chart.js; custom tweet-cleaning pipeline handles URLs, mentions, and hashtags.

Air Quality & Weather Monitoring and Forecasting *Python, Streamlit, Pandas, NumPy, scikit-learn, Plotly, Meteostat API*

- Built a Streamlit dashboard merging synthetic air quality (PM2.5, PM10, NO2) and real Mumbai weather data, performing EDA with correlation analysis between wind speed, humidity, and pollutant levels.
- Trained Linear Regression and Random Forest models for pollutant forecasting; predicted graphs closely matched actual pollution trends.

FlavourMap – Indian Food Dataset EDA & Visual Analytics *Python, Pandas, Matplotlib, Seaborn, Power BI, Google Colab*

- Performed full EDA on 6,871 Indian recipes (Kaggle) – including custom time-string parsing, null removal, and column standardisation – generating 6 Python charts and an interactive Power BI dashboard.

Certifications & Courses

- **AWS Academy Graduate – Cloud Foundations** | Amazon Web Services · April 2026 · 20 hours
- **Python Essentials 1** | Cisco Networking Academy / OpenEDG Python Institute · April 2025
- **Networking Basics** | Cisco Networking Academy · December 2024
- **Introduction to Cybersecurity** | Cisco Networking Academy · July 2024
- **Programming Essentials in C (CLA)** | Cisco Networking Academy · April 2024
- **Game Development Using AI** | CogniScience Club, CSI-SAKEC & SAKEC-ACM · March 2025
- **Design Thinking Workshop** | Entrepreneurship & Innovation Cell, SAKEC

Achievements & Activities

- Fullstack Fusion Web Development Hackathon Participant (2024) – CogniScience Club & SAKEC AI&DS
- Co-authored research paper titled "*Digital Legal Rights Awareness Portal to Empower Students Against Cyber Threats in India*" (Submitted, under revision)
- Built 10+ academic and personal software projects across web, cloud, and ML domains
- Maintained a GPA above 8.0 throughout B.Tech